

Predicting oxygen therapy need in COVID-19 patients using a machine learning model



AI Declaration

Susannah Blundell¹, Rob Treharne¹

¹ University of Liverpool, Liverpool, UK

Github:

Email: S.R.Blundell@Liverpool.ac.uk



INTRODUCTION

Silent hypoxia was a critical issue in COVID-19 patients during the 2020 pandemic (Tobin, Laghi and Jubran, 2020), as was ventilator allocation due to limited supply. With the rise of machine learning methods in clinical research, we aim to predict ventilator need in COVID-19 patients using serum protein quantities from mass spectrometry and accompanying patient metadata.

Other models have predicted similar needs in COVID-19 patients, using patient health data (Saadatmand *et al.*, 2022) or a wide variety of clinical and biochemical data (Yamanaka *et al.*, 2022) to predict oxygen therapy needs. For the latter, results showed that of their top predictive features, 3 of 8 were serum proteins, which supports the need for oxygen therapy prediction based on mass spectrometry data as we do here in this preliminary model.

AIMS

PRIMARY AIM

Develop a robust model that scores well on a variety of evaluation metrics

- Rigorous data handling to extract all meaningful information from the data
- Fine-tuning of model development stages such as advanced feature selection, an ensemble of models, and imputation techniques
- External validation with other datasets

METHODS & RESULTS

DATA

The data contains information on hospitalised COVID-19 patients from a hospital in Surrey, containing general health information/clinical measurements ('metadata') and mass spectrometry data of serum proteins.

METADATA	O2	MS DATA
70	1	1033
28	1	518
7	1	266

Fig. 1: Representation of feature count initially, after filtering for missingness, and after feature selection

Model	F1 Score		Accuracy	
	Train	Test	Train	Test
Random Forest	1.00	0.86	1.00	0.83
Gradient Boosting	1.00	0.86	1.00	0.89
Logistic Regression	0.94	0.86	0.94	0.89
SVM	0.99	0.83	0.99	0.85
AdaBoost	1.00	0.77	1.00	0.80
KNN	1.00	0.77	0.94	0.82

Fig. 4: F1 and accuracy scores for train and test data for each simple mode, ranked by test F1 score. Random forest and gradient boosting were carried to the next stage of development.

	TRUE		
	+	-	
PREDICTED	+	11	2
	-	5	17

Fig. 5: Confusion matrix and evaluation metrics for the final random forest model.

Test accuracy: 0.80
Test F1: 0.83

DEVELOPMENT STAGES

PREPROCESSING

DATA

DATA CLEANING

IMPUTATION

ENCODING

SCALING

FEATURE SELECTION

TRAINING

TEST MODELS

TUNING

TRAINING

TESTING

PREDICTION

EVALUATION

METHODS

Processed in pandas – Python 3.11

Filtered to 30% missingness

Multivariate Imputation by Chained Equations with miceforest

Ordinal and binary categories were label encoded

Scaled with scikit-learn StandardScaler

0-variance thresholding and native scikit-learn feature selection methods (see GitHub)

SVM, random forest, AdaBoost, Gradient Boosting, XGBoost, KNN

Hyperopt hyperparameter tuning

Training of final model within MLFlow

Predict using reserved 20% of data

Extract model metrics and plot with matplotlib

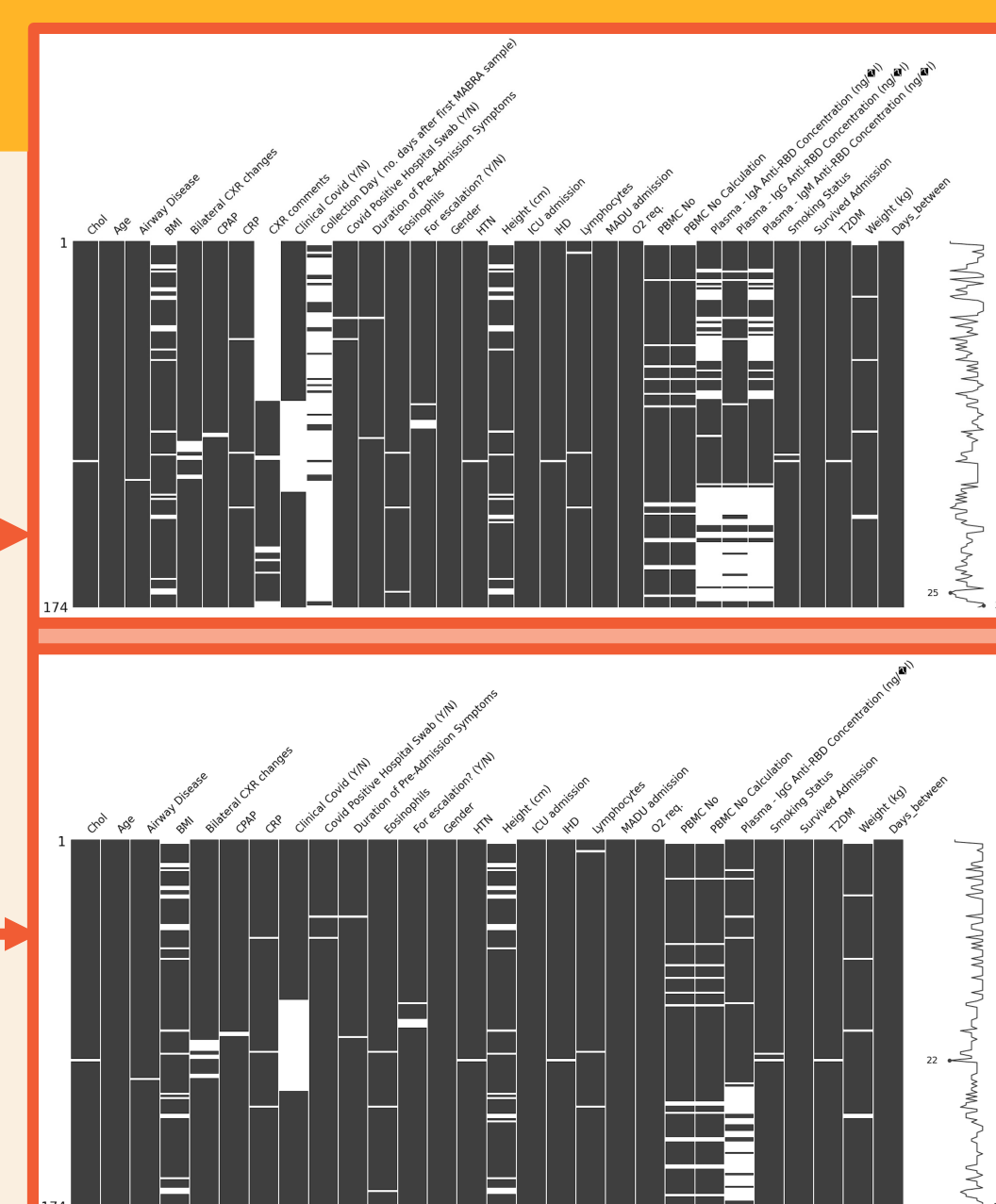


Fig. 2: 'missingno' package representation of missing metadata before and after filtering.

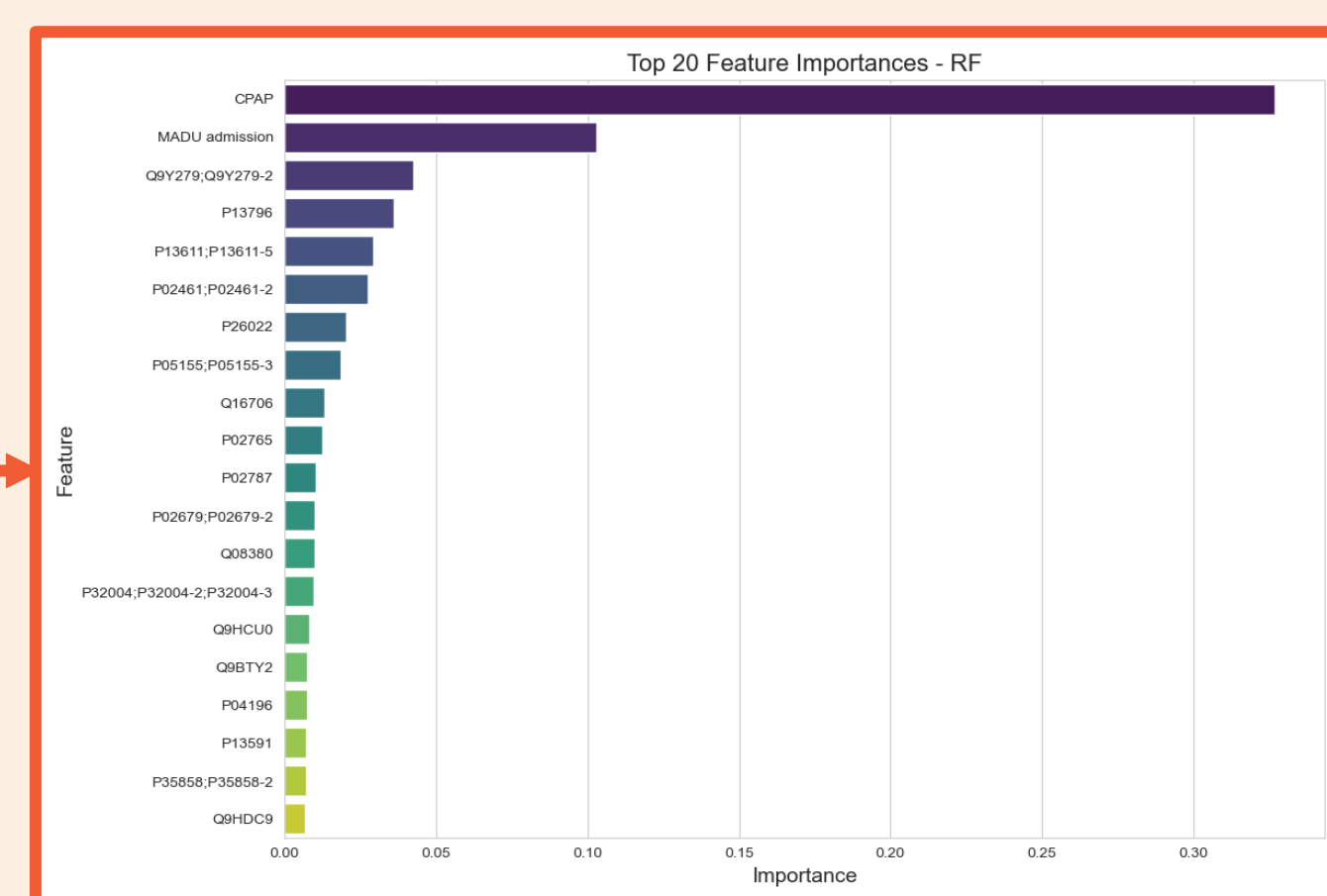


Fig. 3: Bar plot of the top 20 feature importances for the final random forest model, with CPAP, MADU admission and Q9Y279;Q9Y279-2 (a regulator of T-cell proliferation and IL2 production) as the top three features.

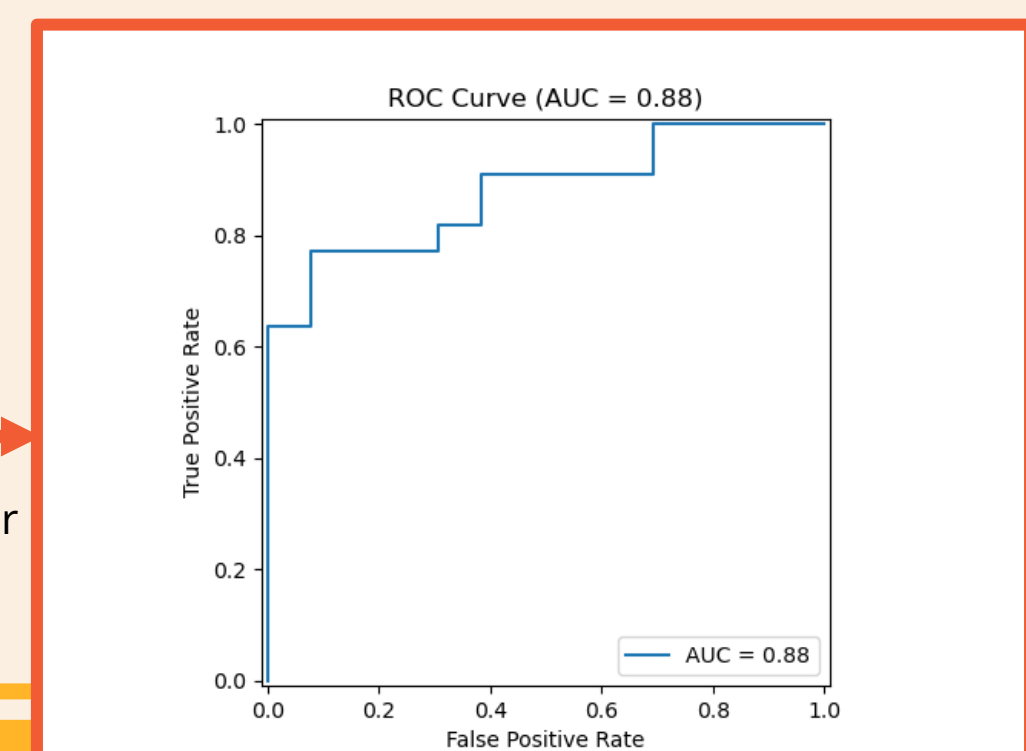


Fig. 6: ROC curve for the final random forest model.

CONCLUSIONS

MAIN RESULTS

- Preliminary model developed is a random forest model
- Currently at 80% accuracy
- Most important features includes two metadata features and the remaining are protein data which is promising evidence for the importance of serum proteins.

FUTURE STEPS

- Further fine tuning; improve data extraction, feature selection, hyperparameter tuning, use model ensemble, etc.
- Incorporate timepoint data with models such as recurrent neural networks
- Apply to other datasets

REFERENCES: Saadatmand, S. *et al.* (2022) 'Predicting the necessity of oxygen therapy in the early stage of COVID-19 using machine learning', *Medical & Biological Engineering & Computing*, 60(4), pp. 957–968. Available at: <https://doi.org/10.1007/s11517-022-02519-x>.

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